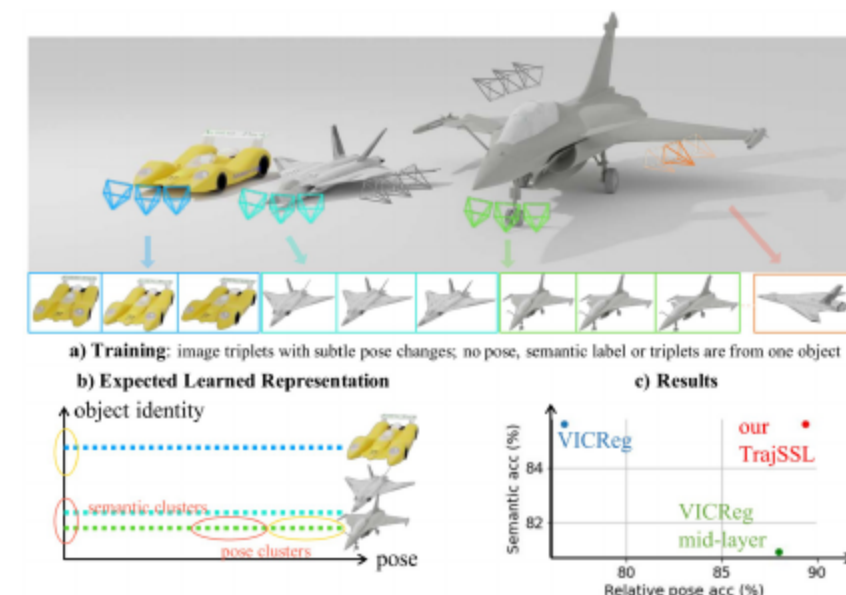


Pose-Aware Self-Supervised Learning with Viewpoint Trajectory Regularization

Jiayun Wang, Yubei Chen, Stella X. Yu

01 Problem Formulation

- Goal: learn features capturing both object identity and pose without labels
- Pose-aware SSL supports navigation, manipulation, and safety-critical perception
- Representations should enable semantic classification and viewpoint estimation
- Avoid over-invariance that removes geometry-critical information



01 Motivation And Gaps

Method	semantic label	pose label	traj. reg.
Fully-Supervised [40, 58]	✓	✓	✗
Geometry-Supervised [18, 26]	✗	✓	✗
Invariant SSL (baseline) [4, 10]	✗	✗	✗
Trajectory-Regularized SSL (ours)	✗	✗	✓

- SSL benchmarks emphasize semantics, neglecting geometric competence
- Invariance objectives generalize poorly to rare or unseen poses
- Lack of label-free benchmark jointly assessing semantics and pose

02 Related Work And Limitations

Invariant SSL For Semantics

Contrastive and non-contrastive SSL achieve pose-invariant embeddings

- Effective for classification but de-emphasize geometric structure
- Lack protocols testing absolute and relative pose understanding

Geometry-Aware And Equivariant Approaches

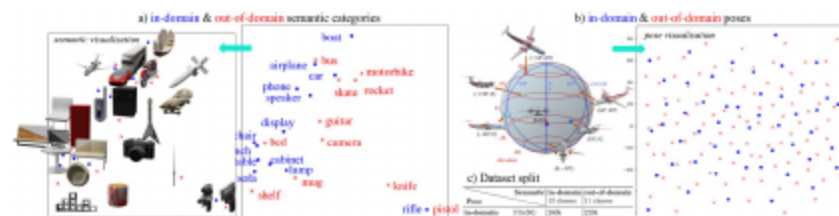
Equivariant/autoencoding methods need group priors or pose supervision

SIE uses ground-truth poses, conflicting with label-free SSL

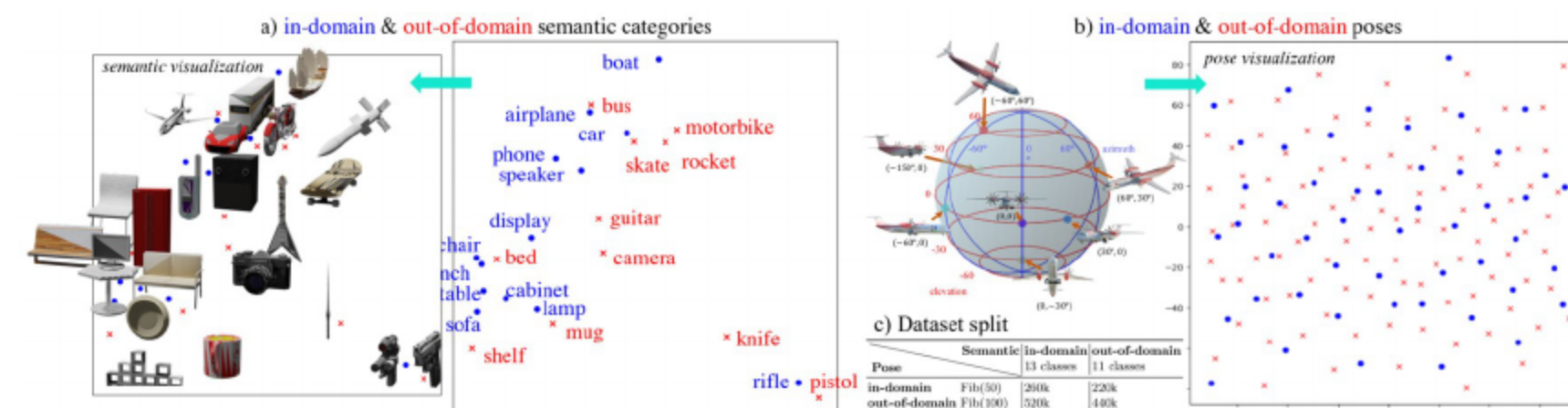
Prior datasets rarely support out-of-domain relative pose tests

02 Pose Estimation Landscape

- Absolute pose requires canonical frames, limiting class-agnostic evaluation
- Relative pose enables out-of-domain tests without canonical dependence
- Fibonacci sphere sampling ensures diverse viewpoints



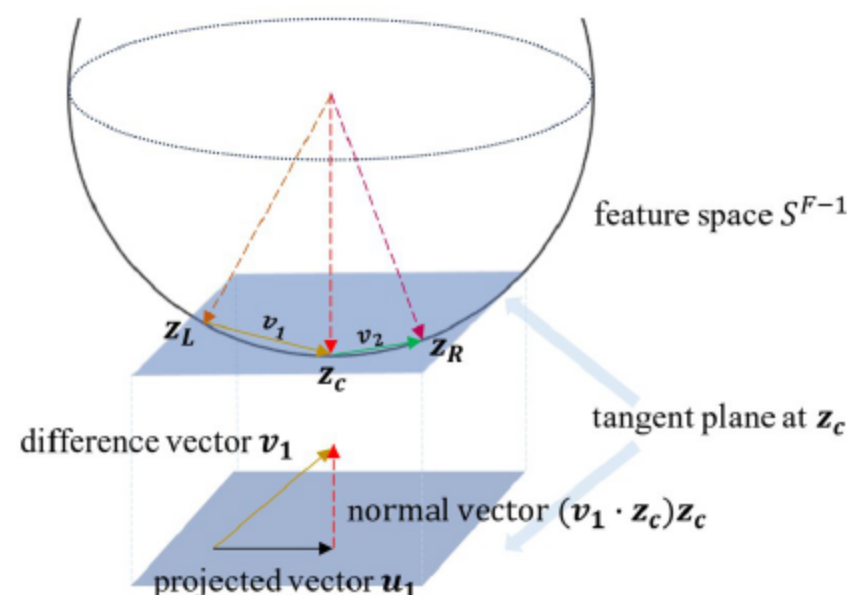
03 Unlabeled Triplet Dataset And Benchmark



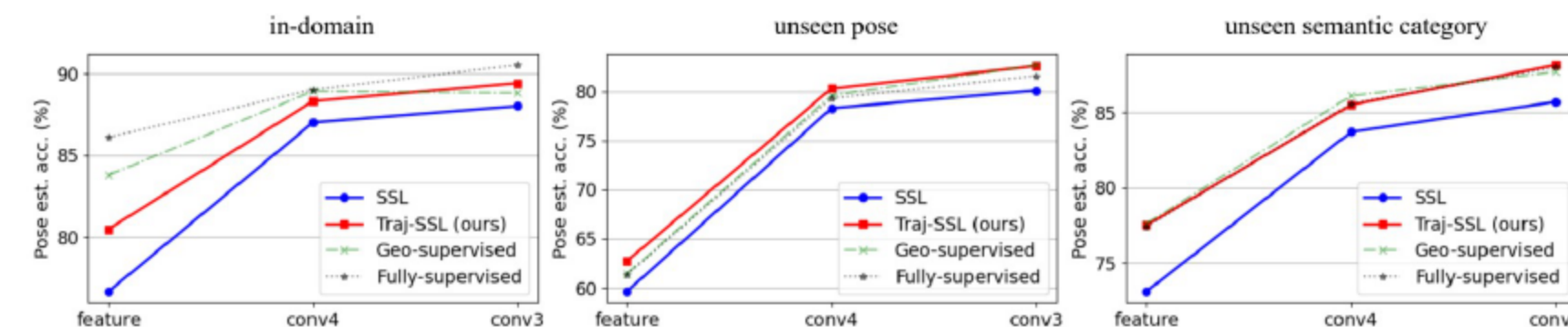
- Introduce triplet dataset from viewpoint trajectories without labels
- Benchmark evaluates semantics and pose, in- and out-of-domain
- Standardized comparison under strict label-free training

03 Viewpoint Trajectory Regularization Loss

- Unsupervised loss enforces locally linear geodesic feature trajectories
- Project differences onto tangent plane and maximize cosine similarity
- Preserves semantic capacity while adding pose awareness

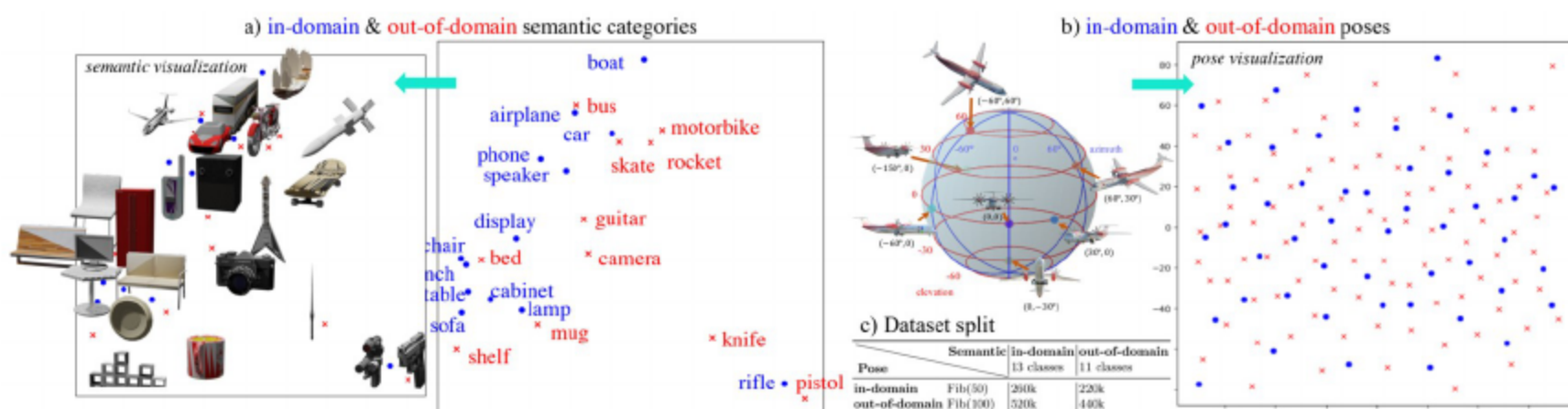


03 Empirical Insights And Gains



- Mid-layer features substantially boost pose estimation over final-layer
- Trajectory regularization adds $\approx 4\%$ pose gain without semantic loss
- Improved generalization to unseen poses, categories, and real images

04 Data Generation And Splits



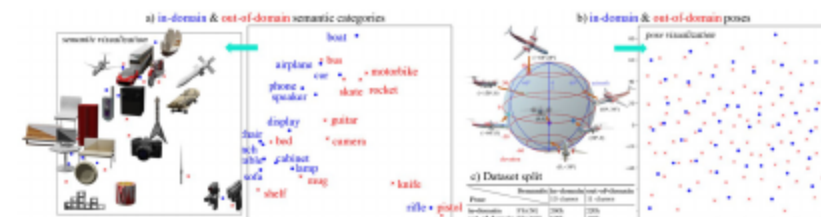
- Render ShapeNet with S2 camera sampling via Fibonacci lattices
- In-domain: Fib(50), 13 categories; OOD: rotated Fib(100), 11 categories
- 400 objects/class; 320 train (unsupervised), 80 test; no shadows

04 Evaluation Scenarios

	Our dataset 3DIEBench 3DIdent		
Out-of-domain evaluation	Yes	No	No
Pose coverage	$(-\pi, \pi)$	$(-\pi/2, \pi/2)$	$(-\pi/2, \pi/2)$
Pose sampling method	even	uneven	uneven
Numer of images	1.5M	2.5M	275k

- Scenarios: in-domain absolute, in-domain relative, OOD poses, OOD categories
- Test global and local pose awareness and generalization

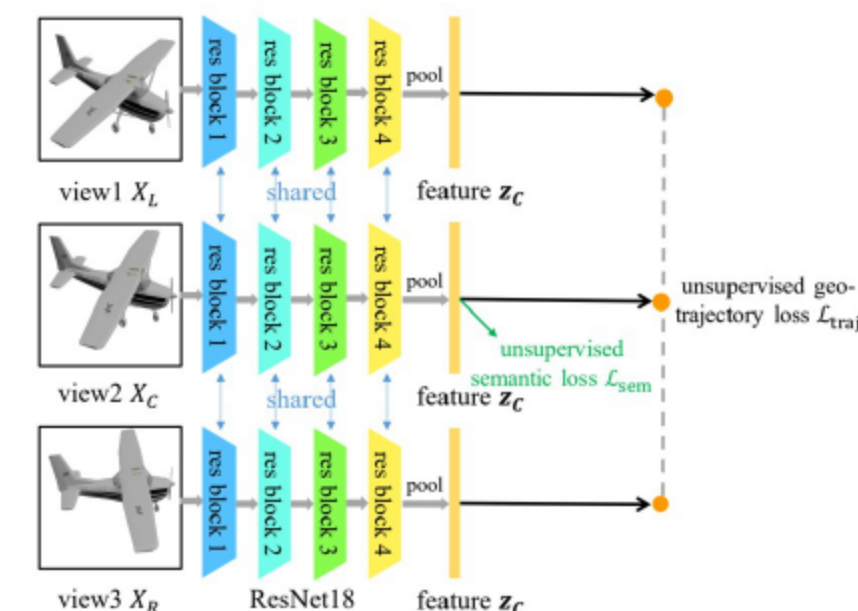
04 Downstream Tasks And Metrics



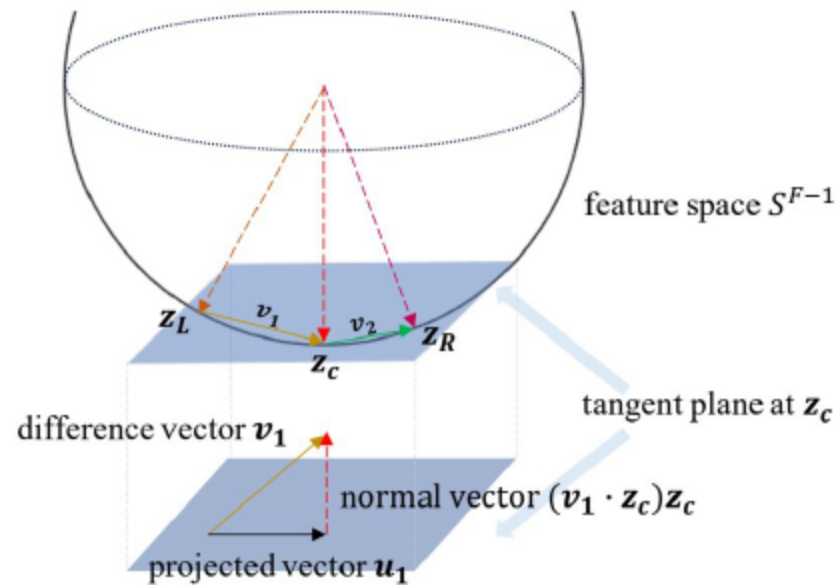
- Linear probe for semantics on frozen features
- Absolute pose: weighted k-NN using canonical frames
- Relative pose: MLP on concatenated pair features, supports OOD

05 Base SSL And Representations

- Start from invariant SSL; treat different poses as distinct samples
- Unit-normalize representations; mid-layers retain geometric cues
- Final-layer targets semantics; mid-layer aids pose estimation



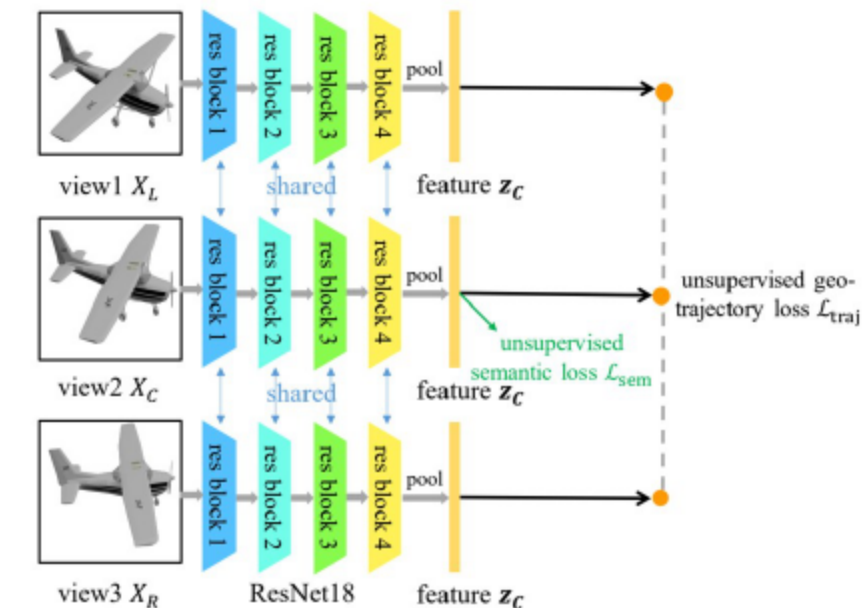
05 Viewpoint Trajectory Regularization



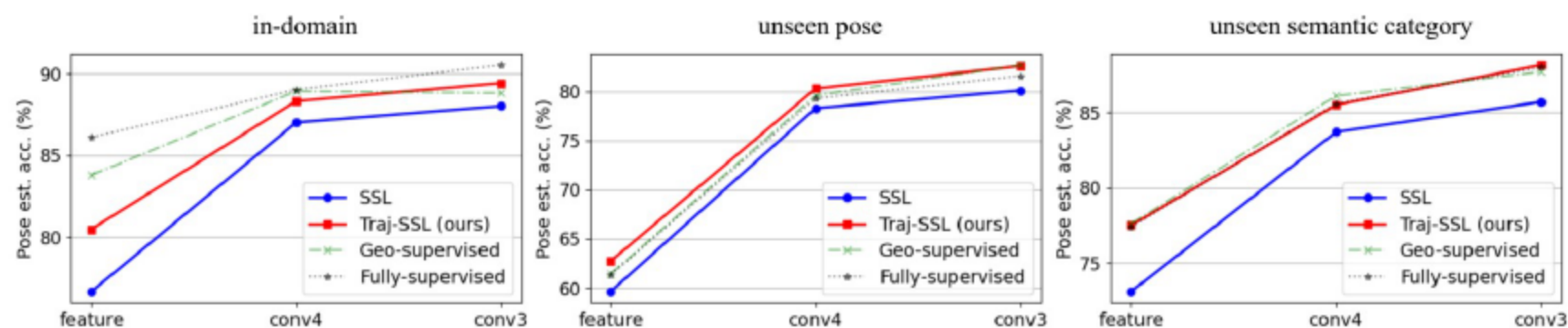
- Use adjacent triplets to align feature differences on tangent plane
- Maximize cosine similarity between projected v_1 and v_2
- Combine with semantic SSL using a balance weight

05 Training Protocol And Data Handling

- ResNet-18, 300 epochs, LARS;
- Adjacent XL; XR via slerp; preserve geometry by avoiding triplet crops
- Apply semantic augmentations only to XC for Lsem

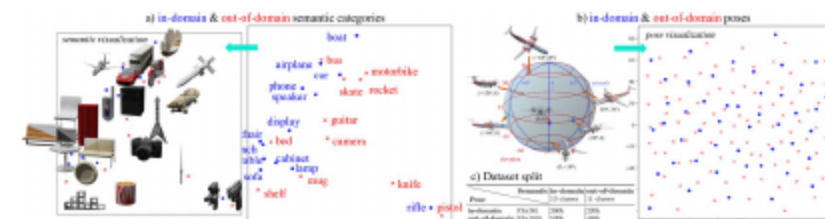


06 Mid-Layer Versus Final-Layer Features



- Mid-layer (conv3/conv4) yields 9–20% pose gains
- Higher-dimensional but compressible with minimal loss

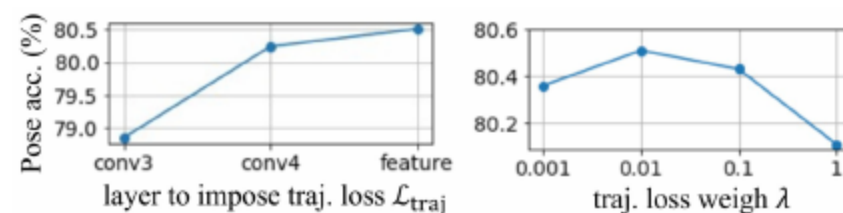
06 Absolute And Relative Pose Estimation



- Absolute: k-NN within categories aligned to canonical pose
- Relative: predict azimuth/elevation deltas via small MLP
- Enables out-of-domain evaluation without canonical frames

06 Loss Placement And Robustness

- Best at pooled feature layer; layer choice changes ~1%
- Robust to non-equidistant triplets and backbone choices
- Maintains semantics while improving geometry

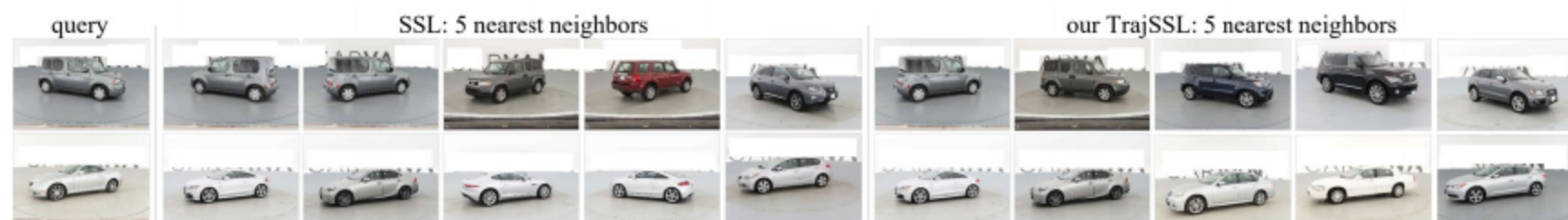


07 Baselines And Training

Acc. (%)	In-Domain					Out-of-Domain Pose Est.				Real Photos	
	sem. cls.	abs. pos	our gain	rel. pose	our gain	unseen sem.	our gain	unseen pose	our gain	Cars [50]	our gain
Fully-Sup. ^a	86.4	92.2		86.1		61.3		77.4		88.5	
Geometry-Sup.	85.4	89.8		83.8		61.4		77.6		87.9	
<i>Fully-unsupervised methods</i>											
VICRreg [4]	85.6	84.3		76.7		59.6		73.1		88.7	
VICRreg+traj.	85.6	87.8	3.5	80.5	3.8	62.7	3.1	77.5	4.4	91.7	3.0
SimCLR [10]	85.9	84.8		77.3		58.1		68.5		89.0	
SimCLR+traj.	86.0	86.4	1.6	79.5	2.2	61.3	3.2	71.0	2.5	91.5	2.5
SimSiam [13]	85.4	84.9		77.4		57.8		68.1		88.8	
SimSiam+traj.	85.5	87.2	2.3	79.5	2.1	61.0	3.2	70.8	2.7	91.2	2.4

- Compare supervised, geometry-supervised, and invariant SSL baselines
- Add trajectory loss to each baseline with identical settings

07 Evaluation Setup



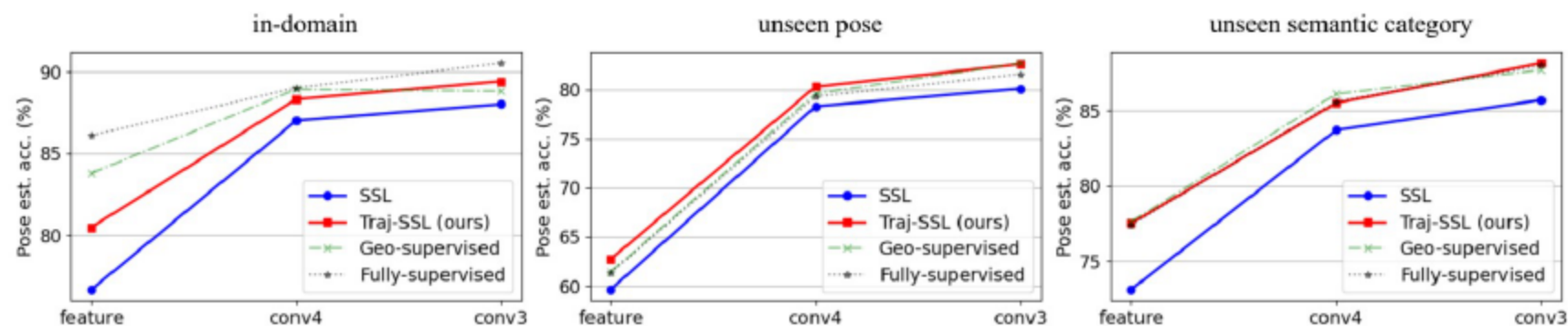
- Semantic linear probe on 512-d; absolute k-NN; relative MLP
- Evaluate in/out-of-domain and on Carvana real data

08 Semantic Classification Performance

Acc. (%)	In-Domain					Out-of-Domain Pose Est.				Real Photos	
	sem. cls.	abs. pos	our gain	rel. pose	our gain	unseen sem.	our gain	unseen pose	our gain	Cars [50]	our gain
Fully-Sup. ^a	86.4	92.2		86.1		61.3		77.4		88.5	
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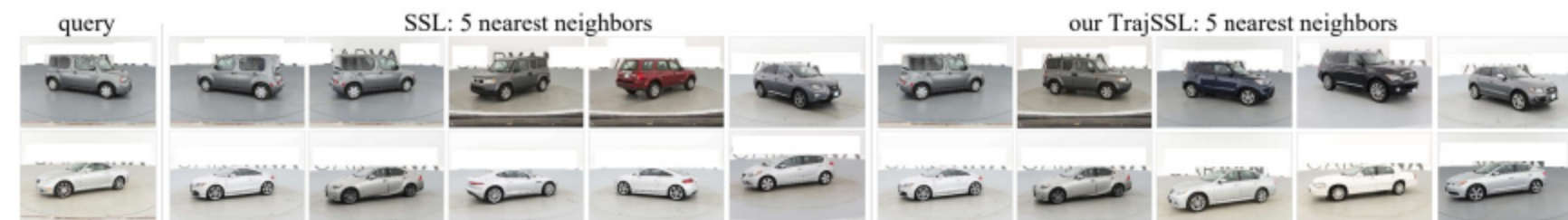
- Semantic accuracy $\approx 85-86\%$ across methods
- Trajectory loss does not harm semantic performance

08 In-Domain Pose Estimation Gains



- Trajectory regularization improves absolute and relative pose by $\approx 4\%$
- Better local linearity and emergent global pose awareness

08 Out-Of-Domain Generalization



- Matches or surpasses supervised on unseen poses/categories
- Carvana: +2–3% and superior pose-similar retrievals

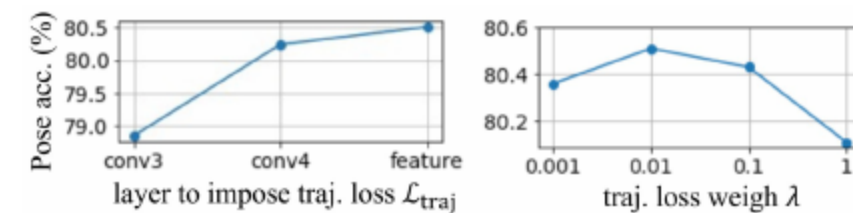
08 Mid-Layer Representation Benefits

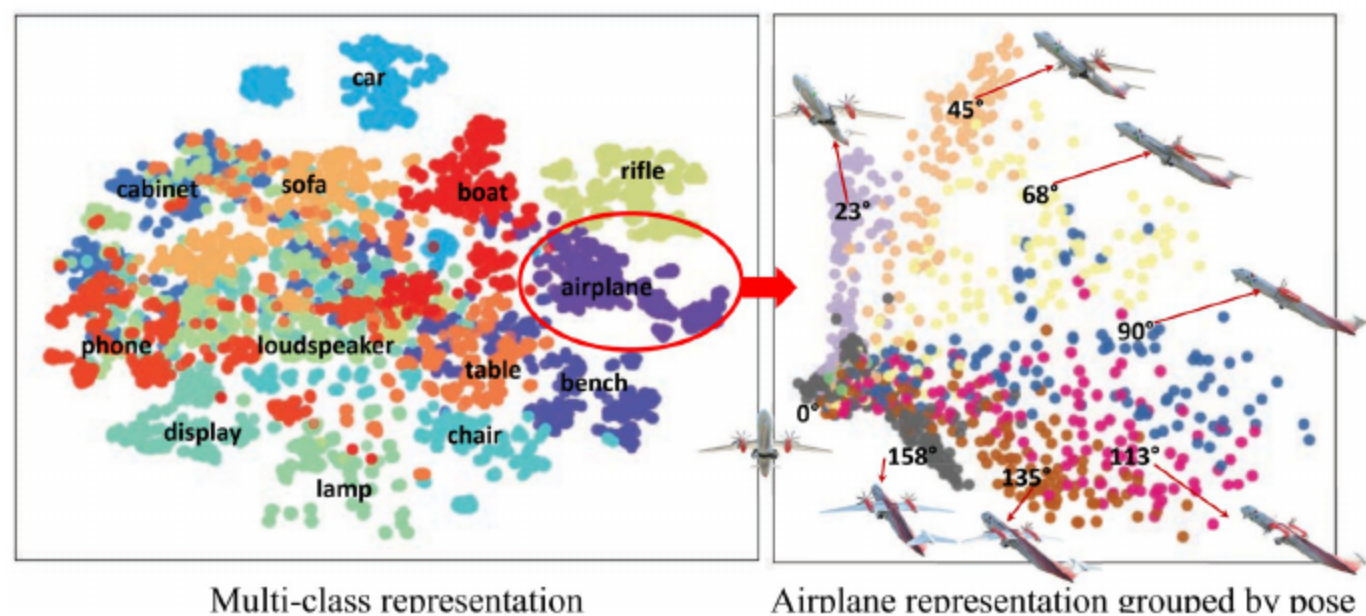
Pose Acc. (%) Representation layer	In-Domain			Unseen Pose			Unseen Cat.		
	conv3	conv4	feat	conv3	conv4	feat	conv3	conv4	feat
Fully-Sup.	90.5	89.0	86.1	81.6	79.3	61.3	88.0	85.6	77.4
Geometry-Sup.	88.8	88.9	83.8	82.7	79.6	61.4	87.7	86.1	77.6
VICReg [4]	88.0	87.0	76.7	80.1	78.3	59.6	85.7	83.7	73.1
VICReg+traj.	89.4 ($\uparrow 9$)	88.3	80.5	82.6 ($\uparrow 20$)	80.3	62.7	88.2 ($\uparrow 11$)	85.5	77.5

- Conv3 boosts: $\approx 9\%$ in-domain, $\approx 20\%$ unseen poses, $\approx 11\%$ unseen classes
- Visualizations show smooth intra-class pose trajectories

09 Ablations On Layers And Hyperparameters

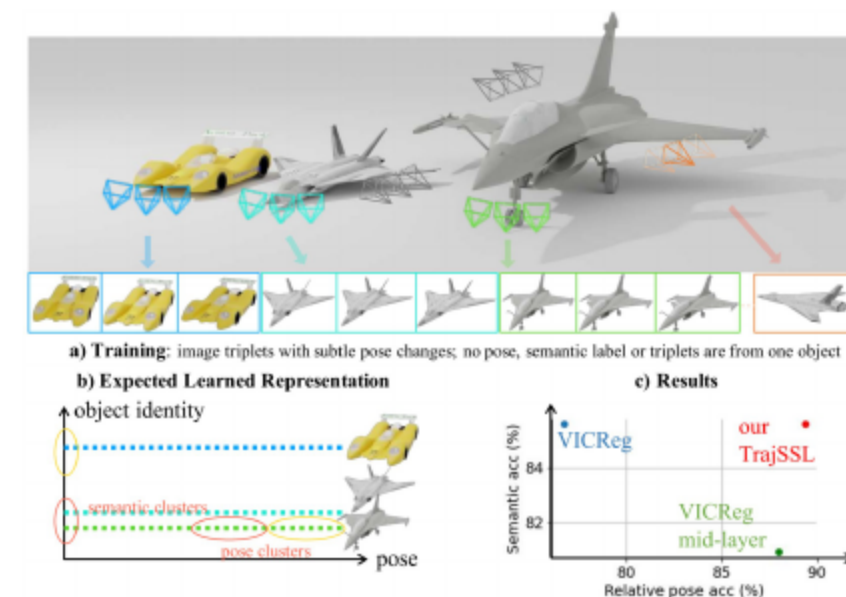
- Layer placement changes $\leq 2\%$; best at feature layer
- Loss weight λ robust; peak near 0.01
- Non-equidistant triplets still yield $\approx 4\%$ gains





- t-SNE/PCA show semantic clusters with smooth pose trajectories
- Baseline lacks coherent geodesic structure

- Label-free benchmark and trajectory regularization elicit pose-aware SSL
- Preserves semantics, improves pose, generalizes across domains



- Limitations: synthetic reliance and focus on 3D orientation
- Future: real trajectories, 6-DoF, depth, flow, stronger priors



THANKS!